Session 4.2: AI Applications in Business Decision-Making Part 1: Predictive Customer Analytics and Churn

TODAY'S PURPOSE

Why This Session Matters

- Use churn predictions to target customers
- Decide who to contact first
- Check whether a retention campaign pays off

Today's flow: 60 min lecture, then 120 min hands-on lab

EXPECTED OUTCOMES

By the end of today, you will be able to:

1. Calculate revenue at risk with Customer Lifetime Value (CLV)
2. Rank customers for retention offers
3. Find the threshold that maximizes campaign Return on Investment (ROI)

Concepts for This Session

CONCEPT 1

Customer Churn Probability

The continuous probability output (predict_proba) generated by a classification model estimating the likelihood of a subscriber canceling services.

CONCEPT 2

Customer Lifetime Value (CLV)

The present value of future net profit margins generated by a customer relationship across their projected relationship tenure.

CONCEPT 3

Revenue At Risk

Quantifying the total annual revenue at risk by multiplying individual churn probabilities by annualized subscriber contract values.

CONCEPT 4

Targeted Retention Incentives

Programmatically matching customer churn tiers to specific cost-effective intervention strategies (e.g., proactive customer support outreach vs. discount incentives).

How the Concept Works in Practice

FOUNDATION 1

The Economics of Retention vs. Acquisition

Acquiring a new customer costs 5 to 7 times more than retaining an existing one. Predictive churn analytics allows subscription businesses to intervene before high-value customers cancel.

FOUNDATION 2

Translating Probability Scores into Financial Exposure

A raw churn probability of 0.85 means nothing to a CEO. Translating that score into $1,800 annual revenue at risk allows executive leadership to evaluate retention budgets quantitatively.

FOUNDATION 3

Targeted Intervention Economics

Do not offer expensive discounts to everyone. Customers with low churn risk will renew anyway (wasted budget); customers with low CLV are unprofitable to save. Machine learning identifies the high-value, high-risk quadrant where retention spend creates maximum impact.

FOUNDATION 4

Evaluating Campaign Return on Investment

Proving that predictive models generate positive financial cash flows. Students calculate Net Retention Benefit: (Saved Customers Annual Value) - (Campaign Cost + Incurred Discounts).

Build, Test, and Diagnose in the IDE

[1] FILE EXPLORER

Organize Assets

Use the session dataset and keep scripts in a clear workspace.

[2] CODE EDITOR

Write Small Segments

Run each completed segment before expanding the program.

[3] AI AGENT CHAT

Diagnose Errors

Capture the traceback, request an explanation, and review the fix.

[4] INTEGRATED TERMINAL

Verify Execution

Confirm the final program completes with return code 0.

Script Demonstration 1: AI Applications in Business Decision-Making Part 1: Predictive Customer Analytics and Churn

```
import pandas as pd
    from sklearn.linear_model import LogisticRegression

    df = pd.read_csv("data/session_04_02_dataset.csv")
    X = df[["Tenure_Months_Count", "Customer_Support_Calls_Count",
"Usage_Drop_Last_30_Days_Pct"]]
    y = df["Churned_Within_60_Days_Binary"]

    model = LogisticRegression().fit(X, y)
    df["Churn_Probability"] = model.predict_proba(X)[:, 1].round(3)
    print(df[["Subscriber_ID", "Tenure_Months_Count",
"Churn_Probability"]].head(3))
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 2: AI Applications in Business Decision-Making Part 1: Predictive Customer Analytics and Churn

```
df["Annual_Subscription_USD"] = df["Monthly_Subscription_Fee_USD"] *
12.0
    df["Revenue_At_Risk_USD"] = (df["Churn_Probability"] *
df["Annual_Subscription_USD"]).round(2)
    total_risk = df["Revenue_At_Risk_USD"].sum()
    print(f"Total Portfolio Revenue at Risk: ${total_risk:,.2f}")
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 3: AI Applications in Business Decision-Making Part 1: Predictive Customer Analytics and Churn

```
def assign_retention_campaign(row):
        if row["Churn_Probability"] >= 0.70 and
row["Annual_Subscription_USD"] >= 1500.0:
            return "VIP_Dedicated_Account_Manager"
        elif row["Churn_Probability"] >= 0.50:
            return "15Pct_Contract_Discount_Offer"
        else:
            return "Standard_Product_Newsletter"

    df["Assigned_Campaign"] = df.apply(assign_retention_campaign,
axis=1)
    print("Retention Campaign Allocation:\\n",
df["Assigned_Campaign"].value_counts())
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Exercise 1 and Exercise 2

EXERCISE 1: GUIDED

Foundational Practice

Follow the lab protocol, run the script, and verify its output with the session dataset.

EXERCISE 2: DEBUGGING

Applied Challenge

Extend the program, diagnose an intentional error, apply the verified correction, and confirm return code 0.